



nextwork.org

Launching VPC Resources



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Create VPC

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances. Mouse over a resource to highlight the related resources.

VPC settings

Resources to create info
Create only the VPC resource or the VPC and other networking resources.

☐ VPC only ☒ VPC and more

Name tag auto-generation info
Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☒ Auto-generate
network

IPv4 CIDR block info
Determine the starting IP and the size of your VPC using CIDR notation.
10.0.0.0/16 65,536 IPs
CIDR block size must be between /16 and /28.

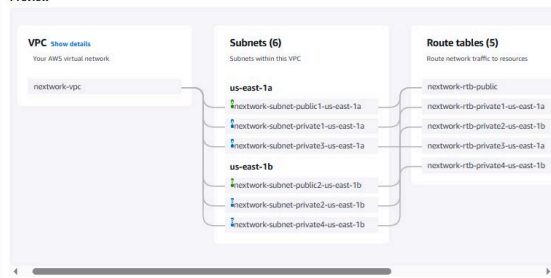
IPv6 CIDR block info
☒ No IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block

Tenancy info
Default

Number of Availability Zones (AZs) info
Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.
1 2 3

▼ Customize AZs
First availability zone

Preview





Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a virtual private network in the AWS cloud that lets me launch resources in an isolated network and control security routing and connectivity.

How I used Amazon VPC in this project

I used Amazon VPC to create public and private subnets launch EC2 instances in each and control how they communicate with each other and the internet.

One thing I didn't expect in this project was...

I didn't expect how quickly the VPC resource map would update and show all connections between subnets route tables and gateways visually.

This project took me...

This project took me 25 minutes.



Setting Up Direct VM Access

It means connecting to my EC2 instance through SSH or RDP so I can control it directly like logging into a computer remotely.

SSH is a key method for directly accessing a VM

SSH means Secure Shell. It is a protocol used to securely connect and communicate with remote servers over a network.

To enable direct access, I set up key pairs

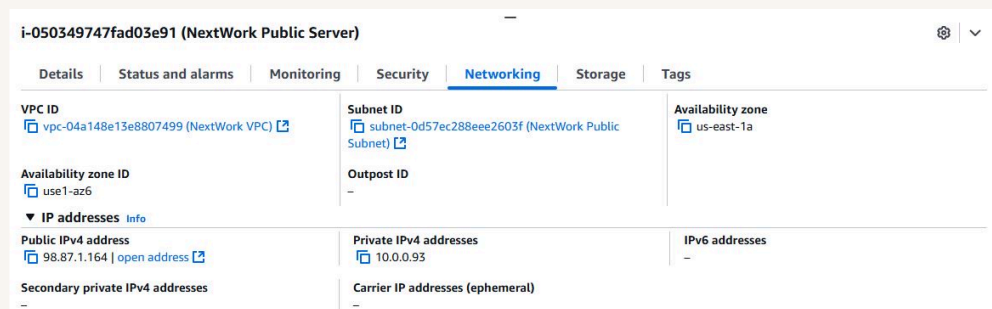
A key pair is a set of security credentials with a public key and a private key that I use to securely connect to my EC2 instance using SSH.

A private key's file format is usually PEM or PPK. My private key was in the PEM format.



Launching a public server

I went to the EC2 console selected my instance then chose Networking and clicked Manage IP Addresses to edit the settings.





Launching a private server

My private server uses a different security group to restrict direct internet access and only allow traffic from the public server for better security.

The new security group's source is the NextWork Public Security Group, allowing only the public server to communicate with the private server.

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, sg-042f6045371b77cd2) Remove

Type Info	Protocol Info	Port range Info
ssh ▼	TCP	22
Source type Info	Source Info	Description - optional Info
Custom ▼	Q Add CIDR, prefix list or security group sg-042f6045371b77cd2 X	e.g. SSH for admin desktop

Add security group rule

► Advanced network configuration

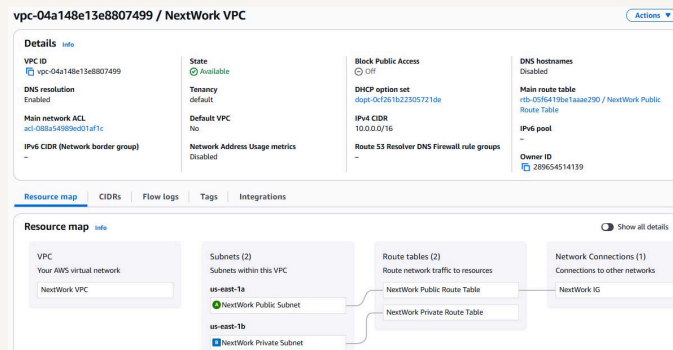


Speeding up VPC creation

I used the VPC wizard to create a new VPC with public and private subnets, route tables, an internet gateway, and enabled name tag auto-generation to organize all resources.

A VPC resource map is a visual diagram that shows how all components like subnets route tables and gateways are connected within a VPC.

The new VPC can have the same IPv4 CIDR block because VPCs are isolated networks and do not conflict unless connected through peering.





Speeding up VPC creation

Tips for using the VPC resource map

I could create up to two public subnets, one for each availability zone, to maintain redundancy and high availability.

NAT gateways allow instances in private subnets to access the internet for updates while keeping their IP addresses hidden and blocking inbound traffic.

The screenshot displays the 'Create VPC' wizard in the AWS Management Console. The left pane shows the 'VPC settings' section with the following configurations:

- Resources to create:** 'VPC and more' is selected.
- Name tag auto-generation:** 'Auto-generate' is checked, with the name 'nextwork' entered.
- IPv4 CIDR block:** '10.0.0.0/16' is entered, with a note that the CIDR block size must be between /16 and /30.
- IPv6 CIDR block:** 'No IPv6 CIDR block' is selected.
- Tenancy:** 'Default' is selected.
- Number of Availability Zones (AZs):** '2' is selected, with a note recommending at least two AZs for high availability.

The right pane shows a 'Preview' of the VPC resource map. It includes a 'VPC' box labeled 'nextwork-vpc', a 'Subnets (6)' box containing two public subnets ('us-east-1a' and 'us-east-1b') and four private subnets, and a 'Route tables (5)' box containing one public route table and four private route tables. Lines connect the VPC to the subnets and the subnets to their respective route tables.